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A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

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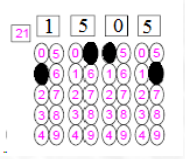
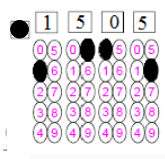
SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 11-01-2026**

Time: 02.00Pm to 05.00Pm **GTM-12** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below

Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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11-01-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main_GTM-12_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. The sum $\sum_{K=1}^{99} \log \sqrt[3]{\frac{K}{K+1}}$ is equal to:

(Assume base of logarithm is 10)

1) 2 **2)** -2 **3)** $\frac{-2}{3}$ **4)** $\frac{2}{3}$

2. If α, β, γ are roots of the equation $x^3 + ax + 10 = 0$ and value of $(\alpha+\beta)^3 + (\beta+\gamma)^3 + (\gamma+\alpha)^3$ is $10k, k \in R$ then k is equal to:

1) 3 **2)** $\frac{3}{2}$ **3)** -3 **4)** $\frac{-3}{2}$

3. If α, β, γ are the positive real roots of the equation $x^3 - 6x^2 + ax + b = 0$ then the minimum value of $\frac{3}{\alpha+1} + \frac{3}{\beta+1} + \frac{3}{\gamma+1}$ is:

1) 1 **2)** 3 **3)** 9 **4)** 27

4. The number of ways in which 4 boys and 4 girls can stand in a circle so that each boy and each girls is one after the other is:

1) 3!.4! **2)** 4! .4! **3)** 8! **4)** 7!

5. When expanded the product $(x+2)(x+3)(x+4)....(x+9)(x+10)$ can be written as $a_9x^9 + a_8x^8 + ...+ a_1x + a_0$. The value of $(a_1 + a_3 + a_5 + a_7 + a_9)$ is:

1) 11! **2)** 11! – 9! **3)** 27 (9!) **4)** $\frac{(11!-9!)}{2}$

6. In triangle ABC, if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$ (where a, b, c are length of the sides of a triangle ABC) , then the value of $\sin^2 A + \cos^2 B + \tan^2 C$ is equal to

1) 2 **2)** 4 **3)** $\frac{9}{2}$ **4)** $\frac{11}{2}$

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7. If at least one child in a family with 3 children is a boy then the probability that exactly 2 of the children are boys, is:

1) 3/7 **2)** 4/7 **3)** 1/3 **4)** 3/8

8. If $Z=(\lambda+3)-i\sqrt{5-\lambda^2}$ then the locus of Z is

1) ellipse **2)** semicircle **3)** parabola **4)** straight line

9. Range of the expression $y=\cos x(\cos x-\cos 3x)$ is equal to:

1) $[0,1]$ **2)** $[0,2]$ **3)** $\left[0,\frac{5}{2}\right]$ **4)** $[0,\infty)$

10. The equations of the sides of triangle are $x+y-5=0, x-y+1=0, and\ y-1=0$ then the coordinates of circumcentre are

1) $(2,1)$ **2)** $(1,2)$ **3)** $(2,-2)$ **4)** $(1,-2)$

11. If a circle passes through the point $(1,2)$ and cuts the circle $x^2+y^2=4$ orthogonally, then the locus of its centre is:

1) $x^2+y^2-2x-6y-7=0$ **2)** $x^2+y^2-3x-8y+1=0$

3) $2x+4y-9=0$ **4)** $2x+4y-1=0$

12. Two mutually perpendicular tangents of the parabola $y^2=4ax$ meet the axis in P_1 and P_2. If S is the focus of the parabola then $\frac{1}{l(SP_1)}+\frac{1}{l(SP_2)}$ is equal to:

($l(AB)$ means distance between A and B)

1) $\frac{4}{a}$ **2)** $\frac{2}{a}$ **3)** $\frac{1}{a}$ **4)** $\frac{1}{4a}$

13. Let E be the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and C be the circle $x^2+y^2=9$. Let P and Q be the points $(1,2)$ and $(2,1)$, respectively. Then:

1) Q lies inside C but outside E **2)** Q lies outside both C and E

3) P lies inside both C and E **4)** P les inside C but outside E

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14. If the chord of the hyperbola $x^2-y^2=a^2$ touches the parabola $y^2=4ax$, then the locus of the middle points of this chord is:

1) $x^3=(x-a)y^2$ **2)** $x^3=(x+a)y^2$ **3)** $x(x^2-y^2)=ay$ **4)** $x^3=x-ay^2$

15. Let $\vec{a}=\hat{i}-\hat{j}, \vec{b}=\hat{j}-\hat{k}$, and $\vec{c}=\hat{k}-\hat{i}$. If $\vec{d}$ is a unit vector such that $\vec{a}.\vec{d}=0$ and $[\vec{b}\ \vec{c}\ \vec{d}]=0$, then $\vec{d}$ is:

1) $\pm\frac{\hat{i}+\hat{j}-\hat{k}}{\sqrt{6}}$ **2)** $\frac{\hat{i}+\hat{j}-2\hat{k}}{\sqrt{6}}$ **3)** $\pm\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{6}}$ **4)** $\frac{\hat{i}-\hat{j}+2\hat{k}}{\sqrt{6}}$

16. Consider the line $\frac{x-3}{3}=\frac{y-2}{1}=\frac{z-1}{0}$, then which of the following statement is incorrect?

1) The line is parallel to xy plane

2) The line is perpendicular to z axis

3) The line passes through point $(6,3,1)$

4) The line intersect the plane $x-3y+z=3$

17. A plane P is perpendicular to the vector $\vec{A}=2\hat{i}+3\hat{j}+6\hat{k}$ and contains the terminal point of the vector $\vec{B}=\hat{i}+5\hat{j}+3\hat{k}$. The distance from the origin to the plane P, is:

1) 1 **2)** 2 **3)** 4 **4)** 5

18. Let $f:D\to R$ defined as $f(x)=\frac{x^2+2x+a}{x^2+4x+3a}$ where D and R denote the domain of f and the set of all real numbers respectively. If f is surjective mapping then the range of a is:

1) $0\le a\le 1$ **2)** $0<a\le 1$ **3)** $0\le a<1$ **4)** $0<a<1$

19. The value of $\tan^{-1}\frac{1}{\sqrt{2}}-\tan^{-1}\frac{\sqrt{5-2\sqrt{6}}}{1+\sqrt{6}}$ is equal to:

1) $\frac{\pi}{6}$ **2)** $\frac{\pi}{4}$ **3)** $\frac{\pi}{3}$ **4)** $\frac{\pi}{2}$

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20. Let $f(x)=\sin x+x$ for all real x.

Statement- 1: The function is strictly increasing on R.

Statement-2 $f'(x)\geq 0$ for all real x.

1) Statement 1is correct statement 2 is correct
2) Statement 1 correct statement 2 is incorrect
3) Statement 1 is incorrect statement 2 is correct
4) Both statements are incorrect

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

21. Find the value of $16\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$

22. The value of $\int_{-1}^{1}\left(x^3+x\right)dx=$

23. Find the number of local extrema of the function $f(x)=x^3-3x^2+1$

24. If $|x^2-7|\leq 9$, then $x\in[\alpha,\beta]$ *where* $\alpha+\beta=$

25. Find the area enclosed by the curve $y=|x|$ and $y=2$ is

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

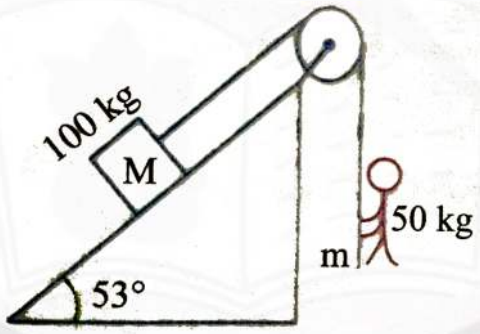
26. A ball is projected horizontally with a speed v from the top of a plane inclined at an angle 45^0 with the horizontal. How far from the point of projection will the ball strike the plane

1) $\frac{v^2}{g}$ **2)** $\sqrt{2}\frac{v^2}{g}$ **3)** $\frac{2v^2}{g}$ **4)** $\sqrt{2}\left[\frac{2v^2}{g}\right]$

27. A swimmer crosses the river along the line making an angle 45^0 with the direction of flow. Velocity the river is $5m/s$. Swimmer takes 6 seconds to cross the river of width 60m.The velocity of the simmer with respect to water will be

1) $10m/s$ **2)** $12m/s$ **3)** $5\sqrt{5}m/s$ **4)** $10\sqrt{2}m/s$

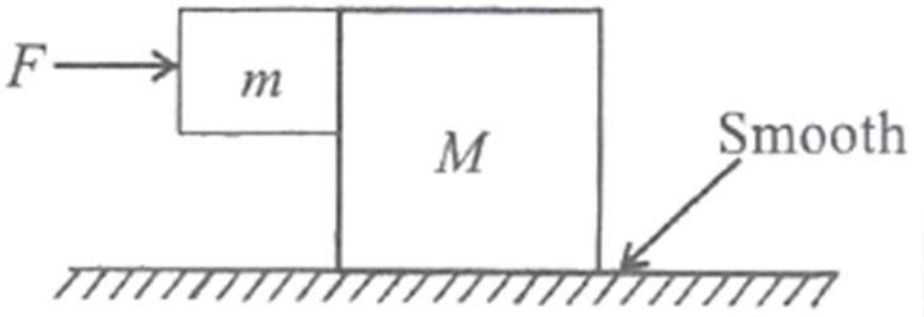
28. In the arrangement shown, by what acceleration the boy must go up so that 100kg block remains stationary on the wedge. The wedge is fixed and friction is absent everywhere. (Take $g=10ms^{-2}$)



1) $2m/s^2$ **2)** $4m/s^2$ **3)** $6m/s^2$ **4)** $8m/s^2$

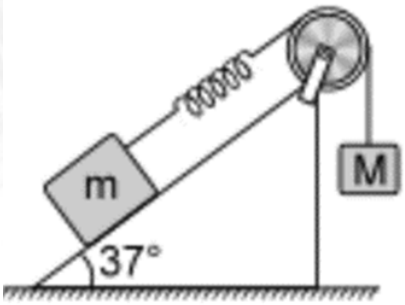
29. The two blocks, $m=10kg$ and M=50 kg are free to move as shown. The coefficient of static friction between the blocks is 0.5 and there is no friction between M and the ground. A minimum horizontal force F is applied to hold m against M that is equal to:

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1) 100 N **2)** 50 N **3)** 240 N **4)** 180 N

30. A block of mass m is attached with a massless spring of force constant k. The block is placed over a fixed rough inclined surface for which the coefficient of friction is $\mu = \frac{3}{4}$. The block of mass m is initially at rest. The block of mass M is released from rest with spring in unstretched state. The minimum value of M required to move the block up the plane is (neglect mass of string and pulley and friction in pulley.)

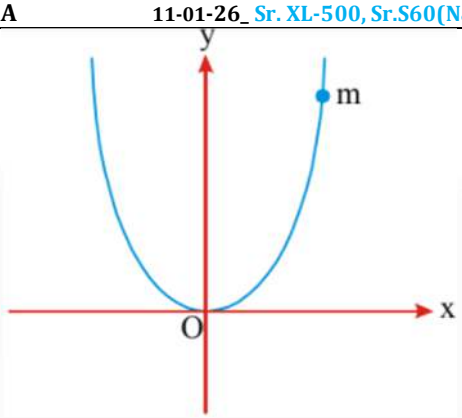


1) $\frac{3}{5}m$ **2)** $\frac{4}{5}m$ **3)** $\frac{6}{5}m$ **4)** $\frac{3}{2}m$

31. A bead of mass m is located on a parabolic wire with its axis vertical and vertex at the origin as shown in figure and whose equation is $x^2=4ay$ .The wire frame is fixed and the bead can slide on it with out friction. The bead is released form the point $y=4a$ on the wire frame from rest. The tangential acceleration of the bead when it reaches the position given by y= a is:

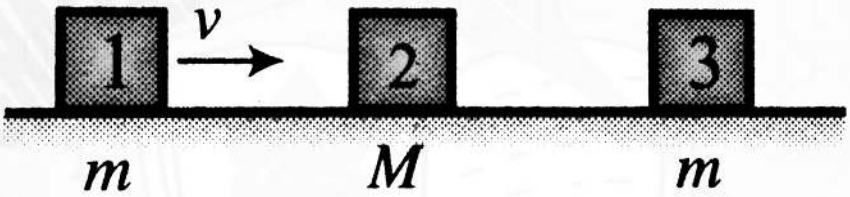
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1) $\frac{g}{2}$ **2)** $\frac{\sqrt{3}g}{2}$ **3)** $\frac{g}{\sqrt{2}}$ **4)** $\frac{g}{\sqrt{5}}$

32. Three blocks are placed on smooth horizontal surface and lie on same horizontal straight line. Block 1 and block 3 have mass m each and block 2 has mass $M\left(M>m\right)$.Block 2 and block 3 are initially stationary, while block 1 is initially moving towards block 2 with speed v as shown. Assume that all collisions are headon and perfectly elastic. What value of $\frac{M}{m}$ ensures that block 1 and block 3 have the same final speed?



1) $5+\sqrt{2}$ **2)** $5-\sqrt{2}$ **3)** $2+\sqrt{5}$ **4)** $3+\sqrt{5}$

33. Statement –I: A sphere of mass m moving with speed u undergoes a perfectly elastic head on collision with another sphere of heavier mass M at rest (M>m), then direction of velocity of sphere of mass m is reversed due to collision (no external force acts on system of two spheres)

Statement –II During a collision of spheres of unequal masses, the heavier exerts more force on lighter mass in comparison to the force which lighter mass exerts on heavier mass

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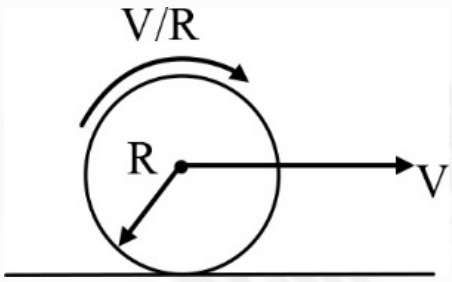
1) Both Statement 1 and Statement 2 are true. statement 2 is correct explanation of statement 1

2) Statement 1 is true but Statement 2 is true. statement 2 is not a correct explanation of statement 1

3) Statement 1 is true but Statement 2 is false.

4) Statement 1 is false and Statement 2 is true

34. A disc is performing pure rolling on a smooth stationary surface with constant angular velocity as shown in figure. At any instant, for the lower most point of the disc



1) Velocity is v, acceleration is zero **2)** Velocity is zero, acceleration is zero

3) Velocity is v, acceleration is $\frac{v^2}{R}$ **4)** Velocity is zero, acceleration is $\frac{v^2}{R}$.

35. Statement- 1: A uniform thin rod of length L is hinged about one of its end and is free to rotate about the hinge without friction. Neglect the effect of gravity. A force F is applied at a distance x from the hinge on the rod such that force always is perpendicular to the rod. As the value of x is increased from zero to L, the component of reaction by hinge on the rod perpendicular to length of rod increases.

Statement- II: Under the conditions given in statement-1 as x is increased from zero to L, the angular acceleration of rod increases.

1) Both Statement 1 and Statement 2 are true. statement 2 is correct explanation of statement 1

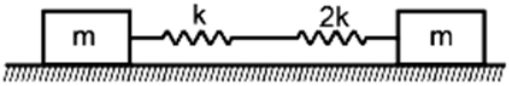
2) Statement 1 is true but Statement 2 is true. statement 2 is not a correct explanation of statement 1

3) Statement 1 is true but Statement 2 is false.

4) Statement 1 is false but Statement 2 is true.

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36. A system is shown in the figure. The time period for small oscillations of the two blocks will be



1) $2\pi\sqrt{\frac{3m}{k}}$ **2)** $2\pi\sqrt{\frac{3m}{2k}}$ **3)** $2\pi\sqrt{\frac{3m}{4k}}$ **4)** $2\pi\sqrt{\frac{3m}{8k}}$

37. Two vibrating strings of same length, same cross section area and stretched to same tension are made of materials with densities ρ & 2ρ.Each string is fixed at both ends. If v_1 represents the fundamental mode of vibration of the one made with density ρ and v_2 for another, then v_1 / v_2 is:

1) $\frac{1}{2}$ **2)** 2 **3)** $\sqrt{2}$ **4)** $\frac{1}{\sqrt{2}}$

38. Match the following for a displacement wave

Column-I	**Column- II**
A) In refraction	p) Speed of wave does not change
B) In reflection	q) Wave length is decreased
C)In refraction from rarer to denser medium	r) Frequency does not change
D) In reflection on a denser medium	s) Phase change of π rad takes place

1) $A \to r; B \to p,r; C \to q,r; D \to p,r,s$

2) $A \to q,s; B \to P,r; C \to s, D \to p,s$

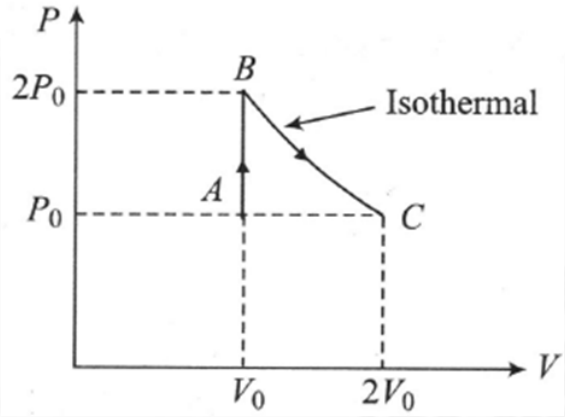
3) $A \to P,r; B \to p,s; C \to q,s; D \to s$

4) $A \to q,s; B \to p,r; C \to p,s; D \to q,s$

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39. A diatomic ideal gas undergoes a thermodynamic change according to the P-V diagram shown in the figure. The total heat given to the gas is nearly



1) $2.5P_0V_0$ **2)** $1.4P_0V_0$ **3)** $3.9P_0V_0$ **4)** $1.1P_0V_0$

40. A dipole of dipole moment $\vec{P} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is placed at point $A(2,-3,1)$. The electric potential due to this dipole at the point $B(4,-1,0)$ is equal to (All the parameters specified here are in S.I. units.)

1) 2×10^9 volts **2)** -2×10^9 volts **3)** 3×10^9 volts **4)** -3×10^9 volts

41. Assertion: When an external resistor of resistance R (connected across a cell of internal resistance r) is varied, power consumed by resistance R is maximum when R=r.

Reason: Power consumed by a resistor of constant resistance R is maximum when current through it is maximum.

1) Both Assertion and Reason are true. Reason is correct explanation of Assertion

2) Assertion is true but Reason is true. Reason is not a correct explanation of Assertion

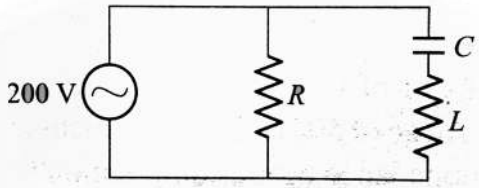
3) Assertion is false but Reason is true.

4) Both Assertion and Reason are false.

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42. In the circuit diagram shown, $X_c = 100\Omega, X_L = 200\Omega$ & $R = 100\Omega$. The effective current through the source is :



1) 2A **2)** $2\sqrt{2}$A **3)** 0.5 A **4)** $\sqrt{0.4}$ A

43. An insulting rod of length l carries a charge q distributed uniformly on it. The rod is pivoted at its midpoint and is rotated at a frequency f about a fixed axis perpendicular to rod and passing through the pivot. The magnetic moment of the rod system is

1) Zero **2)** $\pi q f l^2$ **3)** $\frac{\pi q f l^2}{12}$ **4)** $\frac{\pi q f l^2}{3}$

44. An inductor L and a resistor R are connected in series with a direct current source of emf E. The maximum rate at which energy is stored in the magnetic field is:

1) $\frac{E^2}{4R}$ **2)** $\frac{E^2}{R}$ **3)** $\frac{4E^2}{R}$ **4)** $\frac{2E^2}{R}$

45. Assertion : Two point coherent sources of light S_1 and S_2 are placed on a line as shown, P and Q are two points on that line. If at point P maximum intensity is observed the maximum intensity should also be observed at Q



Reason: In the figure of Assertion, the distance $|S_1P - S_2P|$ is equal to distance $|S_2Q - S_1Q|$

1) Both Assertion and Reason are true. Reason is correct explanation of Assertion

2) Assertion is true but Reason is true. Reason is not a correct explanation of Assertion

3) Assertion is false but Reason is true.

4) Both Assertion and Reason are false.

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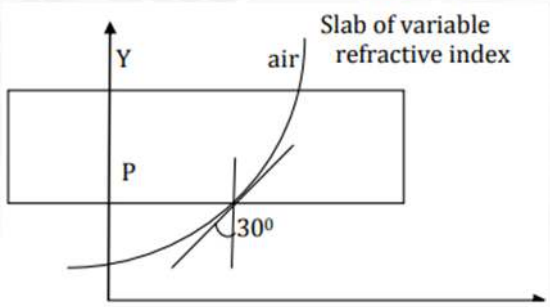
SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

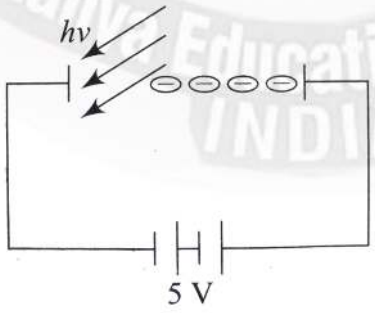
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. A piston of 20 cm diameter and 20 cm long moves down in a cylinder of diameter 20.628 cm. The oil filling the annular space has a viscosity of 10 poise and the weight of the piston is 1 kg. Find the speed with which the piston slides down. $\left[g = 10\,m/\sec^2\right]$ in cm/s

47. A ray of light travelling in air is incident at angle of incident 30^0 on one surface of a slab in which refractive index varies with y. The light travels along the curve $y = 4x^2$ (y and x are in metre) in the slab. If the refractive index of the slab at $y = \left(\frac{1}{2}\right)m$ in the slab in n, find $10n$



48. Photons of energy $5\,eV$ are incident on cathode as shown in the figure. Electrons reaching the anode have kinetic energies varying from $6\,eV$ to $8\,eV$. Find the work function of the metal in eV.



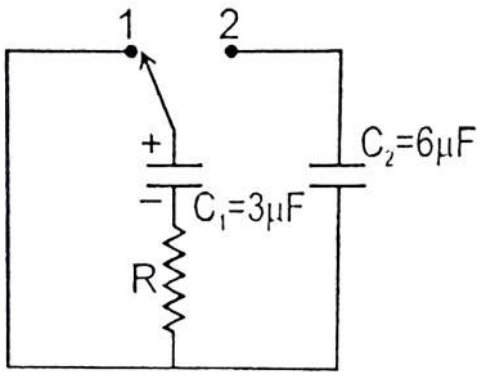
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49. Consider the reaction $N^{14} + \alpha \rightarrow O^{17} + p$

The masses of $N^{14}, He^{4}, H^{1}, O^{17}$ are respectively 14.00307 u, 4.00260 u, 1.00783 u and 16.99913 u. Find the total kinetic energy of the products if the striking α particle has the minimum kinetic energy required to initiate the reaction.(in KeV)

1u =931.5 MeV/c^2

50. A charged capacitor $C_1 = 3\mu F$ is discharged through $R = 1K\Omega$ by putting the switch in position 1. When the current reaches $I_0 = 2A$, the switch is thrown to position 2 to discharge through uncharged capacitor $C_2 = 6\mu F$ and steady state is allowed to reach. Find the heat dissipated (in Joules) in the resistor R after switch is thrown to position 2.



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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. In a first order decomposition reaction, the time taken for the decomposition of reactant to one fourth and one sixteenth of its initial concentration are t_1 and t_2 (s), respectively.

The ratio of $\frac{t_1}{t_2}$ will:

1) $\frac{4}{3}$ **2)** $\frac{3}{2}$ **3)** $\frac{3}{4}$ **4)** $\frac{1}{2}$

52. The atomic number of the element from the following with lowest 1st ionization enthalpy is:

1) 32 **2)** 35 **3)** 55 **4)** 19

53 Which of the following binary mixture does not show the behavior of minimum boiling azeotropes?

1) $H_2O + CH_3COC_2H_5$ **2)** $C_6H_5OH + C_6H_5NH_2$

3) $CS_2 + CH_3COCH_3$ **4)** $CH_3OH + CHCl_3$

54. Correct statement for an element with atomic number 9 are

A) There can be 5 electrons for which $m_s = +\frac{1}{2}$ and 4 electrons for which $m_s = -\frac{1}{2}$

B) There are two electrons in 3s orbital

C) The last electron goes to orbital with n=2 and $l = 1$

D) The sum of angular nodes of all the atomic orbital is 5

Choose the correct answer from the options given below:

1) C and D only **2)** A and C only **3)** A,C and D only **4)** A and B only

55. The number of species from the following that are involved in sp^3d^2 hybridization is

$\left[Cu(NH_3)_4\right]^{2+}, SF_6, \left[CrF_6\right]^{3-}, \left[CoF_6\right]^{3-}, \left[Mn(CN)_6\right]^{3-}$ and $\left[MnCl_6\right]^{3-}$

1) 5 **2)** 6 **3)** 4 **4)** 3

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56. Match the List –I with List –II

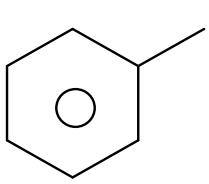
List –I (Reagent)	**List-II (Functional Group detected)**
A. Sodium bicarbonate solution	I. double bond/ unsaturation
B. Neutral ferric chloride	II. Carboxylic acid
C. Ceric ammonium nitrate	III. Phenolic- OH
D. Alkaline $KMnO_4, Cold$	IV. Alcoholic –OH

Choose the correct answer from the options given below:

1) A- II, B-III,C-IV,D-I **2)** A-II, B-III,C-I,D-IV

3) A-III, B-II,C-IV,D-I **4)** A-II,B-IV,C-III,D-I

57. Which of the following amine show positive carbylamine test?

A) 

B) $(CH_3)_2 NH$ C) CH_3NH_2 D) $(CH_3)_3 N$

1) A and C **2)** C only **3)** B only **4)** B and D

58. Statement- I:-In group 14 elements Sn and Pb have the least first ionization enthalpy values (they are lowest among their group members.)

Statement –II:- Sn^{+2} and Pb^{+4} the nature of their ions are reducing and oxidizing respectively

1) Statement I is true but statement II is false

2) Both statement I and statement II are false

3) Statement I is false but statement II is true

4) Both statement I and statement II are true

59. Match the List- I with List –II.

List –I Molecule/ ion		**List-II Bond pair: lone pair (On the central atom)**	
A.	ICl_2^-	I.	4:2
B.	H_2O	II.	2:1
C.	SO_2	III.	2:3
D.	XeF_4	IV.	2 :2

Choose the correct answer from the options given below:

1) $A-IV, B-III, C-II, D-I$ **2)** $A-III, B-IV, C-II, D-I$

3) $A-III, B-IV, C-I, D-II$ **4)** $A-II, B-I, C-IV, D-III$

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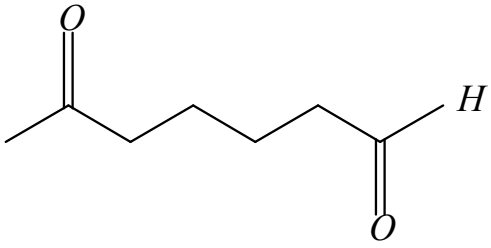
60. Pair of transition metal ions having the same number of unpaired electron is:

1) V^{2+}, Co^{2+} **2)** Ti^{2+}, Co^{2+} **3)** Fe^{3+}, Cr^{2+} **4)** Ti^{3+}, Mn^{2+}

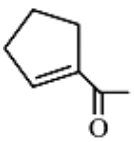
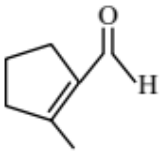
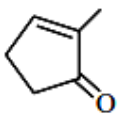
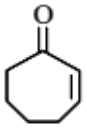
61. Given below are two statements:
Assertion: Alcohols are formed when alkyl chlorides are treated with aqueous potassium hydroxide by elimination reaction.
Reason: In alcoholic potassium hydroxide, alkyl chlorides form alkenes by abstracting the hydrogen from the β – carbon.
In the light of the above statements, choose the most appropriate answer from the options given below.
1) Both Assertion and Reason are true. Reason is correct explanation of Assertion
2) Assertion is true but Reason is false.
3) Assertion is false but Reason is true.
4) Both Assertion and Reason are false.

62. **Assertion:-** The N-N single bond is weaker than that the P-P single bond
Reason:- N-N bond is weaker than P-P bond because of high interelectronic repulsion of non bonding electrons
Choose the correct answer from the options given below
1) Both Assertion and Reason are true. Reason is correct explanation of Assertion
2) Assertion is true but Reason is false.
3) Assertion is false but Reason is true.
4) Both Assertion and Reason are false.

63. Number of molecules from below which CANNOT give iodoform reaction is:
2-Butanone, Butanal, 2- Pentanone, 3- Pentanone, Pentanal and 3- Pentanol
1) 5 **2)** 4 **3)** 3 **4)** 2



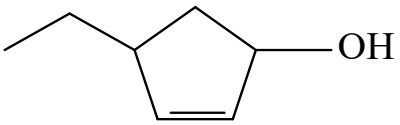
64. When under goes
Intermolecular aldol condensation, the major product formed is:

1)  **2)**  **3)**  **4)** 

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65. What is the correct IUPAC name of



1) 4- Ethyl – 1- hydroxycyclopent-2- ene

2) 1- Ethyl – 3 – hydroxycyclopent – 2 – ene

3) 1- Ethylcyclopent – 2- en-3-ol

4) 4- Ethylcyclopent- 2 –en-1- ol

66. Given below are Two statements:

Statement- I: H_2Se is more acidic than H_2Te.

Statement- II: H_2Se has higher bond enthalpy for dissociation than H_2Te.

In the light of the above statements, choose the correct answer from the options given below.

1) Statement I is true but statement II is false

2) Both statement I and statement II are false

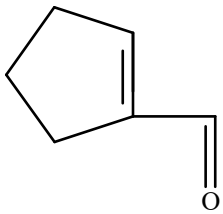
3) Statement I is false but statement II is true

4) Both statement I and statement II are true

67. In SO_2, NO_2^- and N_3^- the hybridizations at the central atom are respectively:

1) sp^2, sp^2 *and* sp **2)** sp^2, sp *and* sp **3)** sp^2, sp^2 and sp^2 **4)** sp^2, sp^2 and sp^3

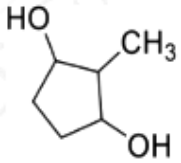
68.

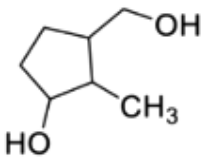


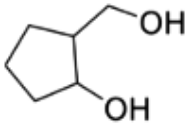
The given compound undergoes following reactions identify final product

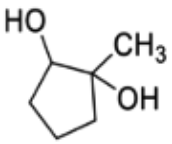
i) Ethylene glycol/ H^+ ii) $(a)BH_3, (b)H_2O_2, NaOH$

iii) H_3O^+ iv) $NaBH_4$

1) 

2) 

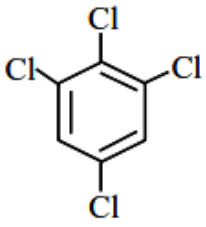
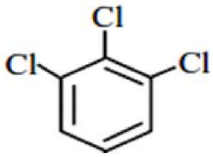
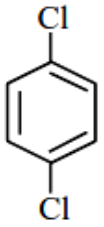
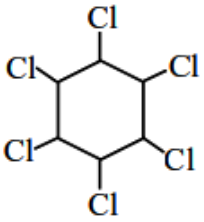
3) 

4) 

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69. In the following reaction sequence, the major product Q is

L- Glucose $\xrightarrow[(ii)Cr_2O_3,775K,10\text{–}20atm]{(i)HI,\Delta} P \xrightarrow[UV]{3Cl_2} Q$

1)  **2)**  **3)**  **4)** 

70. In neutral of faintly alkaline solution, 8 moles of permanganate anion quantitatively oxidize thiosulphate anions to produce sulphate ion. The number of moles sulphate ion is

1) 2 **2)** 4 **3)** 6 **4)** 8

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. Among $[I_3]^+, [SiO_4]^{4-}, SO_2Cl_2, XeF_2, SF_4, ClF_3, Ni(CO)_4, XeO_2F_2, [PtCl_4]^{2-}, XeF_4$, and $SOCl_2$, the total number of species sp^3 hybridised central atom is ____

72. 20 ml sodium iodide solution gave 4.74 g silver iodide when treated with excess of silver nitrate solution. The molarity of the sodium iodide solution is _____M.
(Nearest Integer value) (Given= Na=23, I=127, Ag=108, N=14, O=16 $gmol^{-1}$)

73. The amount of calcium oxide produced on heating 150 kg limestone(75% pure) is _____kg (Nearest integer)

Given : Molar mass (in $g\,mol^{-1}$) of $Ca-40, O-16, C-12$.

74. How many compounds can show tautomerism

Phenol	Acetone	Acetaldehyde	Cyclohexanone

75. The number of paramagnetic metal complex species among $[Co(NH_3)_6]^{3+}, [Co(C_2O_4)_3]^{3-}, [MnCl_6]^{3-}, [Mn(CN)_6]^{3-}, [CoF_6]^{3-}, [Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$ with same number of unpaired electrons is _____.

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